

Who Says Linux Is Not (in) Rocket Science?

People have always looked at the sky and dreamed of flying to the stars. However, throughout history, it was impossible to create a reliable engine that could overcome the Earth's gravity and pull a payload into orbit. Luckily, the scientific achievements in the 20th century—in chemistry, in physics and in the theory of interplanetary flights—have allowed mankind to build rocket engines capable of not only carrying a satellite, but also humans.



Perhaps we all remember the time when the first artificial satellite, Sputnik, was launched into a low altitude elliptical orbit around Earth. That was October 4, 1957. In general, you can look at this date as the beginning of an era of human space flights. Sputnik [1] was a very simple device by today's standards. It could only transmit radio signals (-beep-beep-beep-) at frequencies of 20 and 40 KHz, besides being only 58 centimetres in diameter and weighing around 800 kilograms—a very tiny unit compared to its carrier rocket, R7, which had a height of about 30 metres and weighed 280 tons.

The history of rocket development has roots in ancient times, when gunpowder was invented and when the first fireworks appeared in China. We can't say for sure whether it was clear back then how to create a missile and why, if at all, it ought to be created. However, this firework became the basic element of a cannon—in other words, modern artillery. Cannons used a controlled explosion of gunpowder—the energy that was released pushed a missile towards a certain direction. Afterwards, a whole new science in artillery came up, called ballistics—another fundamental element that helped launch rockets.